

Link lengths

Symbol	Value	Description
L_femur	173.78 mm	Hip pivot → knee pivot (A→C)
L_stub	35.13 mm	Tibia stub, knee upward to coupler end (C→E)
L_tibia	129.39 mm	Knee downward to wheel centre (C→W)
Lc	150.81 mm	Coupler link (F→E)

Total tibia tube length is 164.52 mm (stub + lower), matching the 165 mm tube in the mass comment on [params.py:392](#).

- Relative to the hip motor shaft (A), in the leg plane:

Offset	Value	Direction
$\Delta X = F_x - 0$	-58.87 mm	aft (behind the hip shaft)
ΔY	0 mm	same leg plane, both at ± 143.0 mm
$\Delta Z = F_z - A_z$	+5.29 mm	above the hip shaft

Wheel

Symbol	Value
wheel1_r	75 mm (Ø150 mm)

